

[This question paper contains 4 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 4102

H

Unique Paper Code : 2352012402

Name of the Paper : Multivariate Calculus

Name of the Course : B.A./B.Sc. (H)

Semester : IV (DSC)

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt **all** questions by selecting **two** parts from each question.
3. **All** questions carry equal marks.
4. Use of Calculator not allowed.

1. ~~(a)~~ Let f be the function defined by

$$f(x,y) = \begin{cases} \frac{xy^3}{x^2+y^6}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Is f continuous at $(0,0)$? Explain.

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(b) A cylindrical tank is 4 ft high and has an outer diameter of 2 ft. The walls of tank are 0.2 inches thick. Approximate the volume of interior of tank assuming the tank has a top and a bottom that are both 0.2 inches thick.

(c) Find an equation for each horizontal tangent plane to the surface

$$z = 5 - x^2 - y^2 + 4y.$$

2. (a) Let $f(x, y, z) = ye^{x+z} + ze^{y-x}$. At the point $(2, 2, -2)$ find the unit vector pointing in the direction of most rapid increase of f . And what is the value of most rapid increase of f ?

(b) Find the absolute extrema of the function $f(x, y) = 2 \sin x + 5 \cos y$ on the set S where S is the rectangular region with vertices $(0, 0)$, $(2, 0)$, $(2, 5)$ and $(0, 5)$.

(c) Find the point on the plane $2x + y + z = 1$ that is nearest to the origin.

3. (a) (i) Sketch the region of integration and compute the double integral

$$\int_0^{\pi/2} \int_0^{\sin x} e^x \cos x \, dy \, dx.$$

(ii) Evaluate $\iint_D (2y - x) \, dA$ where D is the region bounded by $y = x^2$, $y = 2x$.

~~(b)~~ Evaluate the area bounded by $r = 2 \cos \theta$.

~~(c)~~ Evaluate the given integral by converting to polar coordinates

$$I = \int_0^2 \int_y^{\sqrt{8-y^2}} \frac{1}{\sqrt{1+x^2+y^2}} dy dx .$$

4. ~~(a)~~ ~~(i)~~ Set up a triple integral to find volume of solid bounded above by paraboloid $z = 6 - x^2 - y^2$ and below by $z = 2x^2 + y^2$.

~~(ii)~~ Set up a double integral to find volume of solid region bounded by ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 .$$

~~(b)~~ Use cylindrical co-ordinates to compute integral

$$\iiint_D z(x^2 + y^2)^{-1/2} dx dy dz \text{ where } D \text{ is the solid}$$

bounded above by plane $z = 2$ and bounded below by surface $2z = x^2 + y^2$.

~~(c)~~ Find volume of solid D where D is the intersection of solid sphere $x^2 + y^2 + z^2 \leq 9$ and solid cylinder $x^2 + y^2 \leq 1$.

5. (a) Evaluate $\oint_C (x^2 y dx - xy dy)$, where C is the path that begins at $(0,0)$, goes to $(1,1)$ along the parabola $y = x^2$, and then return to $(0,0)$ along the line $y = x$.

(b) Prove or disprove that the line integrals are path independent.

(c) Evaluate the line integral

$$\oint_C [(2x - x^2 y)e^{-xy} + \tan^{-1} y]i + \left[\frac{x}{y^2 + 1} - x^3 e^{-xy}\right]j \cdot dR,$$

where C , the curve with parametric equations $x = t^2 \cos \pi t$, $y = e^{-t} \sin \pi t$, $0 \leq t \leq 1$.

6. (a) Find the area enclosed by the semicircle $y = \sqrt{4 - x^2}$ using the line integral.

(b) Find the mass of the homogenous lamina of density $\delta = x$ that has the shape of the surface S given by $z = 4 - x - 2y$ with $z \geq 0$, $x \geq 0$ and $y \geq 0$.

(c) Let $F = 2xi - 3yj + 5zk$, and let S be the hemisphere $z = \sqrt{9 - x^2 - y^2}$ together with the disk $x^2 + y^2 \leq 9$ in the xy -plane. Verify the divergence theorem.